

MUNEEB AHMAD

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<https://mun-ahmd.github.io/apples-and-bananas>

Davis, California
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EDUCATION

University of California, Davis, CA, USA

Sept '25

Master of Science in Electrical and Computer Engineering

- June '27

Coursework: Modelling & Optimization in Computer Engineering, High Performance Statistical Computing, Computer Architecture, Information Theory & Coding

Advisor: John Owens

Jamia Millia Islamia, New Delhi, India

2021

Bachelor of Technology in Computer Engineering

- 2025

CGPA: 9.05 out of 10

Coursework: Computer Networks, Network Security, Computer Architecture, Digital System Design, Embedded Systems, Operating Systems, Parallel & Distributed Computing

SKILLS

- **Languages** - C, C++, CUDA, Triton, Python, Rust, GLSL, HLSL, JavaScript, SQL
- **Frameworks** - PyTorch, Tensorflow, vLLM, Vulkan, DirectX12, OpenGL, OpenMP, MPI, STM32-HAL
- **Tools** - NVIDIA Nsight, PyTorch profiler, RenderDoc, Visual Studio, perf, CMake, Perfetto

WORK EXPERIENCE

Graduate Student Researcher at [Owensgroup](#), UC Davis

Nov '25 -

- Creating optimized GPU kernels for paged attention and integration with vLLM.
- Working on the full DL software stack, from PyTorch to the GPU programs.

Present

Researcher at [BeyondDefence Lab](#), University of New Mexico

Jan '25

- Creating a framework to find the optimal deployment strategy for a given deep learning model on a given edge device. Target use cases include satellites and robotics.
- Achieved a ~400% improvement in latency on the Coral EdgeTPU dev board.
- Worked with hardware accelerators such as Coral EdgeTPU and Nvidia Jetson Nano.
- Applied quantization to models and transformed formats for compatibility with hardware.

- Sept '25

Advanced Application Engineering Intern at Accenture, Gurugram, India

May '24

- Gained knowledge of data security standards, compliance frameworks and AWS cloud.
- Performed cybersecurity audits of 3 organizations as part of third-party risk management.

- July '24

Intern at Nikah Forever, New Delhi, India

Jun '23

- Developed content-based filtering algorithm for a best match recommendation system with user activity driven results to improve over the existing rule-based system.
- Optimized processing in numpy to reduce the memory footprint by 50% to allow for deployment of the recommendation system in a resource constrained environment.

- Aug '23

PROJECTS

SpMV on GPUs using DirectX Work Graphs [Github](#)

July '25

- Implemented load balanced Sparse Matrix Vector Multiplication (SpMV)— an important operation used to accelerate deep learning models to the GPU — using DirectX12 Work Graphs to achieve better performance despite the irregular nature of the application.
- Currently benchmarking against existing methods such as the cuSparse library.

- Oct '25

Vulkan Forward Clustered Renderer [Github](#)

April '23

- Implemented a forward-clustered renderer using Vulkan to handle large numbers of lights.
- Compute shader determines lights relevant to a cluster of space for optimal processing.
- Voxel renderer with compute shader based meshing to reduce draw count by 6 times.
- Focus on easy to modify rendering pipeline.

- July '24

KV Cache Compression with KV Press [Github](#)

Sept '25 -
Dec '25

- Implemented KV cache eviction policy that considers recent attention scores and value vectors.
- Integrated with the kvpress framework
- Benchmarked using lm-eval-harness on Longbench using Mistral-7B-Instruct LLM.